GCSE PHYSICS · TEACHING DECK

Energy Stores & Changes

Deck 4 of 4 — Quick recap & multi-step challenges

AQA 8463 | Edexcel 1PH0 | OCR J249 / J250

Whole-topic exit questions, then three chained questions the way real exam papers ask them



Where we're going — 4 decks



1

Systems, transfers & how to describe them

All tiers

2

Kinetic & gravitational potential energy

All tiers

3

Elastic potential energy & bringing it together

Higher stretch

YOU ARE HERE

4

Quick recap & multi-step challenges

Review

A consolidation deck — no new content. Use it as an exit ticket, or save it for the start of the next session as a warm-up.

QUICK RECAP

Whole-topic exit questions — cover Decks 1–3. Answers on the next slide.

1 Name the eight energy stores.

2 A 1500 kg car travels at 10 m/s. Calculate its kinetic energy.

3 A 4 kg mass is lifted 3 m (g = 9.8). Calculate the GPE gained.

4 Describe the energy transfer when a moving car brakes to a stop.

ANSWERS — PART 1

Tutor reveal

1

Kinetic, gravitational potential, elastic potential, thermal, chemical, nuclear, electrostatic, magnetic.

2

½ × 1500 × 10^2 = 75 000 J (75 kJ).

3

4 × 9.8 × 3 = 117.6 J.

4

Kinetic store empties → by heating (friction in the brakes) → thermal store of the brakes and surroundings.

CHALLENGE 1

① *Easy*

A 0.5 kg ball is raised 2 m (g = 9.8) and then dropped. Ignoring air resistance, find its speed as it lands.

METHOD

- GPE gained = m g h = 0.5 × 9.8 × 2 = 9.8 J, all converted to KE.
- Rearrange ½ m v^2 = 9.8 for v.

WORK IT THROUGH

Answer: v = √(2 × 9.8 ÷ 0.5) = √39.2 ≈ 6.3 m/s. (Or v = √(2 g h) = √(2 × 9.8 × 2).)

Mark scheme: *1 for GPE = 9.8 J · 1 for ½mv^2 = 9.8 · 1 for v ≈ 6.3 m/s.*

CHALLENGE 2

② *Medium*

A 1200 kg car travelling at 20 m/s brakes to rest. All its kinetic energy transfers to the thermal store of the brakes. Find the energy transferred and name the pathway.

METHOD

- Write the equation: $E_k = ½\ m\ v^2$
- Substitute: $E_k = ½ \times 1200 \times 20^2 = 600 \times 400$.

WORK IT THROUGH

Answer: E_k = 240 000 J (240 kJ), transferred by heating (friction) to the thermal store of the brakes and surroundings.

Mark scheme: *1 for equation · 1 for substitution · 1 for 240 000 J · 1 for the pathway: by heating.*

CHALLENGE 3

③ *Hard*

A spring (k = 800 N/m) is compressed 0.05 m and launches a 0.02 kg ball straight up. Assuming all the elastic energy becomes GPE, find the maximum height (g = 9.8).

METHOD

- Find the elastic energy: $E_e = ½ \times 800 \times 0.05^2 = 400 \times 0.0025$.
- At the top all of it is GPE: $m\ g\ h = E_e$.
- Rearrange for h: $h = E_e \div (m\ g)$.

WORK IT THROUGH

Answer: $E_e = 1.0$ J, so $h = 1.0 \div (0.02 \times 9.8) = 1.0 \div 0.196 \approx 5.1$ m.

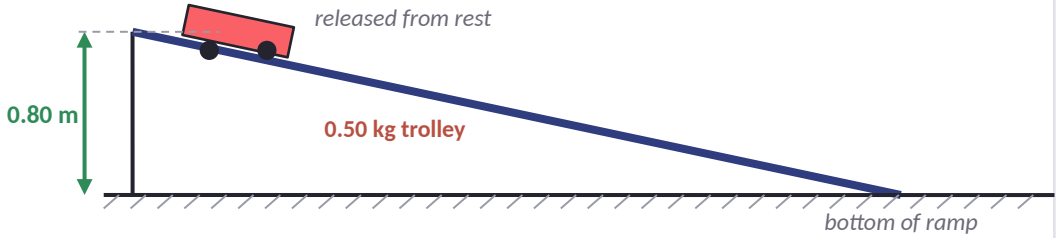
Mark scheme: *1 for $E^e = 1.0$ J · 1 for $m\ g\ h = E^e$ · 1 for rearranging · 1 for h ≈ 5.1 m.*

EXAM PRACTICE — SET A

Chained calculations · 16 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q1 *Diagram · chained calculation* **[8 marks]**

A trolley of mass 0.50 kg is released from rest at the top of a ramp. g = 10 N/kg.



(a) Calculate the GPE lost by the trolley at the bottom of the ramp. [2]

(b) Calculate the speed at the bottom if no energy is dissipated. [3]

(c) The measured speed is 3.6 m/s. Calculate the energy dissipated. [3]

Q2 *Chained calculation · unit conversion* **[8 marks]**

A spring-loaded toy (spring constant 400 N/m) is compressed by 5.0 cm and fires a 10 g ball. All of the elastic energy is transferred to the kinetic store of the ball. g = 10 N/kg.

(a) Calculate the elastic potential energy stored in the spring. [3]

(b) Calculate the speed at which the ball leaves the toy. [3]

(c) The ball is now fired vertically upwards at this speed. Calculate the maximum height it reaches, ignoring air resistance. [2]

MARK SCHEME — SET A

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q1

[8 marks]

M1 (a) $E_p = m\,g\,h = 0.50 \times 10 \times 0.80$

A1 4.0 J

M1 (b) $\frac{1}{2}\,m\,v^2 = E_p$ (ECF from (a))

M1 $v^2 = 2 \times 4.0 \div 0.50 = 16$

A1 4.0 m/s

M1 (c) $E_k = \frac{1}{2} \times 0.50 \times 3.6^2 = 3.24$ J

M1 energy dissipated = 4.0 − 3.24 (ECF)

A1 0.76 J (transferred by heating to the thermal store of the ramp, trolley and surroundings)

Answer: (a) 4.0 J (b) 4.0 m/s (c) 0.76 J

Q2

[8 marks]

B1 (a) 5.0 cm = 0.050 m

M1 $E_e = \frac{1}{2}\,k\,e^2 = \frac{1}{2} \times 400 \times 0.050^2$

A1 0.50 J

B1 (b) 10 g = 0.010 kg

M1 $v^2 = 2E_k \div m = 2 \times 0.50 \div 0.010 = 100$ (ECF from (a))

A1 10 m/s

M1 (c) m g h = 0.50 (all KE → GPE) so h = 0.50 ÷ (0.010 × 10)

A1 5.0 m

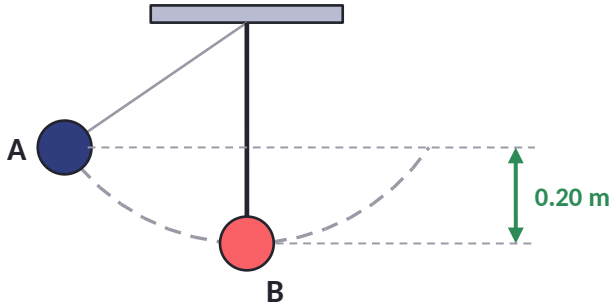
Answer: (a) 0.50 J (b) 10 m/s (c) 5.0 m

EXAM PRACTICE — 6-MARK QUESTION

Levels-based · 6 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q3 *Extended response · levels · uses data* **[6 marks]**

*A pendulum bob of mass 0.050 kg is pulled aside and released from rest at A, which is 0.20 m above the lowest point B (g = 10 N/kg). Describe the energy transfers as the bob swings from A to B and up the other side, and explain why the swing gradually dies away. Use calculations to find the gravitational potential energy at A and the speed at B if no energy were dissipated.



A: released from rest
B: lowest point of the swing
bob of mass 0.050 kg

MARK SCHEME — 6-MARK QUESTION

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q3

[6 marks]

Level 3 (5–6) correct transfers A→B→other side with stores named, the dissipation explanation, AND both calculations (GPE = 0.10 J, v = 2.0 m/s) used to support the answer.

Level 2 (3–4) transfers and one of dissipation or the calculations done well, or both partly.

Level 1 (1–2) isolated points, e.g. “GPE becomes KE”, or one correct calculation only.

Indicative content: at A, GPE = m g h = 0.050 × 10 × 0.20 = 0.10 J and KE = 0; A→B gravitational potential store empties, kinetic store fills; at B all 0.10 J is kinetic, so v = √(2 × 0.10 ÷ 0.050) = 2.0 m/s (or v = √(2 g h)); B→the other side, kinetic empties and GPE fills again. Air resistance and friction at the pivot dissipate energy by heating to the thermal store of the air and surroundings on every swing, so the bob reaches a lower height each time until it stops.

Max 6 marks

Answer: **Levels-based: GPE = 0.10 J, v = 2.0 m/s**

Watch out — and what to remember

IN A NUTSHELL

- All three equations rearrange the same way — write it, substitute, then solve line by line.
- Chained questions are just single steps stitched together — find the intermediate value first.
- Method marks are awarded even when the final answer is wrong — insist on every line of working.

COMMON SLIP-UPS

Rushing straight to a final answer

On a multi-step question, write down each intermediate result (e.g. the energy, before the height) — it's where most of the marks sit.

Losing track of units mid-chain

Carry units through every line, especially when a result from one equation feeds into another.

Skipping the assumption

“Ignoring air resistance” or “all the energy converts” should be stated, not just used silently.

NEXT Part 2 — Heating, power, efficiency & energy resources →